

The FIN Kernel as an Unknown Mathematical Object

An independent referee report on the classification, spectral structure,
and adaptive-information interpretation of the operators

$K_{\text{legacy_ont}}$ and $K_{\text{strict_gate}}$
of the Fractal-Nadsoliton-Theory repository

Research-level referee report with independent numerical verification

Prepared on the basis of repository state HEAD (2026-07-17)

July 2026

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Abstract and verdict table

Scope. This report analyzes the mathematical core of the Fractal-Nadsoliton-Theory (FIN) repository — two frozen kernel functions $K_{\text{legacy_ont}}$ and $K_{\text{strict_gate}}$ used as coupling operators on a 12-node cycle and on random Euclidean point clouds — stripped of all physical interpretation. The guiding question is whether the kernel is more naturally interpreted as an *adaptive information-propagation operator* than as a physical field theory. All spectral and numerical claims in the repository's own audits, and in a previous independent analysis (repo file `Analiza_operatorajadra_FIN.md`, 2026-07-16), were re-derived from scratch; nothing was taken on trust.

Principal results. (i) On its operational domain $\mathbb{Z}_{12}$, $K_{\text{strict_gate}}$ is, to a Frobenius residual of 2.9%, a **normal-ordered discrete Yukawa (screened-Poisson) Green function** $\mathcal{N}[A(L + m^2 I)^{-1}]$ of the cycle Laplacian, with $A = 2.053$, $m^2 = 0.749$ — not a "smoothed Helmholtz operator," as the previous analysis conjectured, and not "the operator rather than its Green function," as that analysis concluded; the correct statement is the reverse. (ii) $K_{\text{legacy_ont}}$ is a **band-pass spectral-shell filter** — a smeared, non-idempotent analogue of the spectral measure $\delta(\mu - \mu_*)$ of the Laplacian — confirming, with one correction, the "on-shell" reading of the previous analysis. (iii) Neither kernel is a reproducing (Mercer) kernel, a Gaussian-process covariance, a transition operator, or a pure resolvent; the classification battery is exhaustive at the 7-mode resolution of the problem. (iv) As an adaptive object, the coupled system $\dot{\psi} = iK_t\psi$, $\dot{K} = \eta(\psi\psi^\top - \gamma K)$ is well-posed and stable only with decay; **the kernel is provably not a fixed point of self-driven Hebbian learning** (correlation of the learned operator with the time-averaged covariance of the system's own excitations is 1.000; with the kernel itself, -0.07). (v) The four kernel parameters are **not jointly identifiable** from the kernel's action: the spectral Fisher matrix has a numerically exact null

direction, and a manifestly distinct parameter tuple reproduces the operational profile to 6×10^{-5} . (vi) On random Euclidean clouds the kernel matrices are Euclidean random matrices with Wigner–Dyson level statistics ($\langle r \rangle = 0.51$), bulk modes localized on ~ 4 –5 sites, and — for the strict kernel — a heat-kernel spectral dimension $d_s \approx 3$ at intermediate times, recovering the embedding dimension. (vii) FIN has, unintentionally, rediscovered a known object: the Chung–Yau discrete Green function (screened variant) on the cycle, resp. a spectral-shell filter; the novelty is not the object but an **unstudied combination**: a normal-ordered oscillatory Green kernel used simultaneously as Hamiltonian, as graph filter, and as the slow variable of a self-referential Hebbian plasticity loop. That combination supports one honest publication; it does not support a new paradigm.

Table 0. Verdict table. Confidence: **H** $\geq 95\%$, **M** 80–95%, **L** 50–80%.

Claim	Verdict	Confidence
K_{strict} = normal-ordered discrete Yukawa Green function on $\mathbb{Z}_{12}$	True, $\ R\ _F / \ K\ _F = 0.029$, $m^2 = 0.749$	H (99%)
K_{legacy} = smeared spectral-shell (band-pass) filter, on-shell object	True; corr. with shell filter 0.950; not idempotent, hence not a projector	H (97%)
Previous analysis: " $K_{\text{strict}} \approx c(z - L)e^{-tL}$, closer to the operator than to its Green function"	Refuted : rational-cutoff/resolvent+contact fit is 5 $\times$ better (nRMSE 0.025 vs 0.121); operator/Green assignment inverted	H (97%)
Repo claim: " $K(d)$ looks like a damped-Helmholtz Green function"	False in every tested regime (residuals 4.8–6.7, repo's own QW-2139; continuum FT sign-indefinite)	H (98%)
Repo claim: Hebbian learning "generates" the kernel (nadsoliton_neural_analysis.py)	False as stated: rule learns the drive covariance (corr 0.9998 with bare cosine), not the kernel	H (99%)
Adaptive interpretation of K is mathematically coherent	True, with decay/homeostasis; fixed points are projector mixtures, never K itself	H (95%)
Four parameters $(\omega, \phi, \beta, \eta)$ identifiable from kernel action	False: Fisher matrix singular; effective dimension ≤ 3 on the domain	H (96%)
FIN unintentionally rediscovers a known operator	Yes: Chung–Yau screened discrete Green function (strict); spectral-shell filter (legacy)	H (95%)
Physics-free publishable contribution exists	Yes, one paper (operator/bootstrap analysis); journal targets in §10	M (85%)
Natural home of the bare kernel	Spectral graph theory / graph signal processing; see §11 ranking	H (95%)

1 Provenance, exact definitions, and the convention trap

1.1 What the repository actually contains

The FIN repository (21,348 files, 564 MB) is a monorepo of an automated theory-development program. Its mathematical content reduces, for the present purpose, to two frozen parametric kernels and a dynamics built on

them. The definitions below were extracted from the primary sources, not from secondary summaries:

$$K_{\text{legacy_ont}}(d) = \alpha_{\text{geo}} \frac{\cos(\omega d + \phi)}{1 + \beta_{\text{tors}} d}, \quad \alpha_{\text{geo}} = 4 \ln 2 = 2.772589, \quad \omega = \frac{\pi}{4}, \quad \phi = \frac{\pi}{6}, \quad \beta_{\text{tors}} =$$

from `TOE_FINAL_DOCUMENTATION_RELEASE_4_4_LEGACY_FULL.tex` (Eq. 408 therein) and (1-1) `RELEASE_6_2_SHANNON_VOID_ASYMMETRY_EN_PL.md`, §4.1; and

$$K_{\text{strict_gate}}(d) = \frac{\cos(\omega d + \phi)}{1 + \beta d^\eta}, \quad \omega = 0.18575, \quad \phi = 0.1625, \quad \beta = 1.0, \quad \eta = 1.8,$$

from the machine-readable gate artifact `report_qw2118_ktotal_spectral_tripartition_gate.json` (1-2) (kernel: `omega=0.18575`, `phi=0.1625`, `beta=1.0`, `eta=1.8`), which also fixes the matrix specification: `n_octaves=12`, `diagonal: zero_self_coupling`, `distance: cyclic_octave_distance`. The strict kernel is therefore the operator $K_{ij} = K_{\text{strict}}(d_{ij})$ on the cycle $\mathbb{Z}_{12}$ with

$$d_{ij} = \min(|i - j|, 12 - |i - j|) \in \{1, \dots, 6\}, \quad K_{ii} = 0.$$

The dynamics used across the repository (e.g. `QW-593_Information_Unity.py`) is the linear flow (1-3)

$$\dot{\psi} = i K \psi, \quad \psi(t) = e^{i K t} \psi_0,$$

i.e. K plays the role of a **Hamiltonian** (a continuous-time quantum walk generator), not of a propagator; the (1-4) propagator is $e^{i K t}$.

1.2 The convention trap — four different operators hide under one name

A referee's first duty is to note that "the FIN kernel" is not one object. The repository uses, interchangeably and often silently, at least four distinct operators built from Eq. (1-1)–(1-3):

Table 1-1. The four operator variants found in the repository and their consequences.

Variant	Where used	Structure	Consequences
Real, cyclic distance, $d \leq 6$	QW-2118 gate, strict-core chain	Symmetric circulant, diagonalized by DFT	Real spectrum, degenerate doublets; the object analyzed here
Real, absolute distance, $d \leq 11$	<code>nadsoliton_neural_analysis.py</code>	Symmetric Toeplitz (path graph)	No circulant symbol; row sums not constant; boundary effects
Complex, $e^{i(\omega d + \phi)}$	QW-593	Non-Hermitian Toeplitz on random clouds	iK not anti-Hermitian $\Rightarrow$ dynamics (1-4) is not unitary; authors renormalize ψ by hand each step
Real, on random Euclidean clouds	QW-540–593 family	Euclidean random matrix (ERM)	Spectrum is a random variable; see §6

Unless stated otherwise, every statement below refers to variant 1 on $\mathbb{Z}_{12}$, which is the only one with a frozen, gate-certified definition. I verified the strict spectrum against the repository's own JSON artifact: all twelve eigenvalues agree to nine significant digits ($\lambda_{\min} = -0.68187476$, $\lambda_{\max} = 1.66030728$; artifact: `report_qw2118...json`).

1.3 Method of this report

Every quantity was recomputed independently: kernels were rebuilt from Eqs. (1-1)–(1-3); spectra, fits, learning experiments and random-matrix statistics were produced by fresh code (protocols in Appendix A). Where the previous analysis (repo file `Analiza_operatora_jadra_FIN.md`) is quoted, it is treated as a *claim to be falsified*, per the task. Confidence levels are attached to every major conclusion (**H** ≥ 95%, **M** 80–95%, **L** 50–80%).

2 Part 1 — Identification of the mathematical object

2.1 Exact spectral symbols

Both kernels are circulants on $\mathbb{Z}_{12}$, hence diagonalized by the discrete Fourier basis $v_m(j) = e^{2\pi i m j / 12} / \sqrt{12}$. Writing the cycle-Laplacian eigenvalues as $\mu_m = 2 - 2 \cos(2\pi m / 12)$, the kernel symbols are

$$\lambda_m = 2 \sum_{d=1}^5 K(d) \cos\left(\frac{2\pi m d}{12}\right) + K(6)(-1)^m, \quad m = 0, \dots, 6,$$

with multiplicities 1, 2, 2, 2, 2, 2, 1. Table 2-1 gives the exact symbols (recomputed; they coincide with (2-1) the previous analysis's table to all printed digits).

Table 2-1. Exact symbols of the two kernels on $\mathbb{Z}_{12}$, against cycle-Laplacian eigenvalues μ_m and wavenumbers $k_m = 2\pi m / 12$.

m	k_m	μ_m	λ_m (strict)	λ_m (legacy)	sign pattern
0	0.000	0.000	+1.6603	−11.1741	+ / −
1	0.524	0.268	+0.9062	+2.0562	+ / +
2	1.047	1.000	+0.0833	+10.2030	+ / +
3	1.571	2.000	−0.3011	−3.2072	− / −
4	2.094	3.000	−0.5393	−0.2516	− / −
5	2.618	3.732	−0.6383	−2.7725	− / −
6	3.142	4.000	−0.6819	−0.8819	− / −

Three structural facts are immediate from the definitions, before any fitting:

Proposition 2.1 (elementary obstructions).

- (a) Neither kernel is positive semidefinite: $\lambda_{\min}(K_{\text{strict}}) = -0.6819$, $\lambda_{\min}(K_{\text{legacy}}) = -11.1741$. Hence neither is a Mercer kernel, an RKHS kernel, or a Gaussian-process covariance matrix. The correct habitat for indefinite kernels is a reproducing-kernel Kreĭn space^{[26][27]}.
- (b) K_{legacy} cannot be a monotone matrix function $g(L)$ of the cycle Laplacian for any monotone g : its symbol is non-monotone in μ (two sign changes, $- + + - - -$). In particular it is not a resolvent $(zI - L)^{-1}$, not a heat kernel e^{-tL} , not any completely monotone function of L .
- (c) K_{strict} is elementwise non-negative on its domain (minimum off-diagonal entry $+0.01107$): the first zero of $\cos(\omega d + \phi)$ occurs at $d_{\star} = (\frac{\pi}{2} - \phi)/\omega = 7.58 > 6$, outside the operational domain. Hence, despite spectral indefiniteness, K_{strict} is a legal weighted-graph adjacency matrix, and Perron–Frobenius applies: $\lambda_{\max} = 1.6603$ is simple, with the strictly positive uniform eigenvector $v_0 = \mathbf{1}/\sqrt{12}$ (constant row sums). **[H, EXACT]**

2.2 The classification battery

I fitted every standard operator family to the two symbols. The honest quality metric is the normalized RMS error $\text{nRMSE} = \|\hat{\lambda} - \lambda\|_2 / \|\lambda - \bar{\lambda}\|_2$ over the seven distinct modes (the previous analysis's "max relative error 8.8%" turns out to be measured relative to $|\lambda|_{\max}$, which flattens small-mode errors; on the worst small mode their own fit errs by 77%).

Table 2-2. Family classification battery. Best-fit parameters by least squares; nRMSE and correlation over the 7 distinct modes. Winners in bold.

Candidate family $g(\mu)$	Interpretation	strict: nRMSE / corr		legacy: nRMSE / corr	
$a/(z - \mu)$	pure resolvent (Green function of $L - z$)	1.000	-0.91	0.762	0.76
$a e^{-t\mu}$	heat/diffusion kernel	0.514	0.97	0.718	0.71
$a/(\mu + z)$	Yukawa-type (shifted resolvent)	0.593	0.92	—	—
$a\mu^s$	fractional power of L	0.658	0.78	—	—
$c(z - \mu)e^{-t\mu}$	Helmholtz operator + heat cutoff (prev. analysis)	0.121	0.993	0.999	0.08
$c(z - \mu)e^{-t\mu^2}$	Helmholtz + Gaussian cutoff	0.221	0.975	—	—
$c(z - \mu)/(1 + t\mu)$	Helmholtz + rational cutoff	0.025	0.9997	—	—
$A/(\mu + B) + C$	resolvent + contact (identity) term	0.025	0.9997	—	—
$a e^{-((\mu-b)/s)^2}$	smear spectral shell (band-pass)	—	—	0.763	0.81
polynomials deg 2–3	generic spectral filter	0.195	0.98	0.60	0.80

The two rows in bold are not competitors but the same object: the Möbius identity

$$c \frac{z - \mu}{1 + t\mu} = \frac{A}{\mu + B} + C, \quad A = \frac{c}{t} \left(z + \frac{1}{t} \right), \quad B = \frac{1}{t}, \quad C = -\frac{c}{t}, \quad (2-2)$$

shows that "Helmholtz operator with rational cutoff" is "shifted resolvent plus a constant (contact) term." The best-fit parameters are

$$A = 2.0528, \quad B \equiv m^2 = 0.7492, \quad C = -1.0896, \quad \text{nRMSE} = 0.025.$$

2.3 The strict kernel is a normal-ordered Yukawa Green function

(2-3)

The constant C in Eq. (2-3) is not a nuisance: it is exactly accounted for by the FIN convention of zeroing the diagonal. On a vertex-transitive graph the diagonal of any resolvent is a constant, $G_{ii} = \frac{1}{12} \sum_m A/(\mu_m + B) = 1.0884$; the FIN convention $K_{ii} = 0$ therefore subtracts precisely a multiple of the identity. Numerically $|C| = 1.0896$ matches $A G_{00} = 1.0884$ to 0.1%. Hence:

Theorem 2.2 (identification of $K_{\text{strict_gate}}$; verified numerically, H 99%).

On $\mathbb{Z}_{12}$, with L the combinatorial cycle Laplacian and $\mathcal{N}$ the operation of deleting the diagonal (normal ordering),

$$K_{\text{strict}} = \mathcal{N}[A(L + m^2 I)^{-1}] + R, \quad A = 2.053, \quad m^2 = 0.749, \quad \frac{\|R\|_F}{\|K\|_F} = 0.029,$$

i.e. **the strict kernel is, to 97% in Frobenius norm, the normal-ordered discrete screened-Poisson(2-4) (Yukawa) Green function of the cycle** — a screened instance of the Chung–Yau discrete Green function^[1]. Equivalently, in spectral space $\lambda_m \approx A/(\mu_m + m^2) - A G_{00}$. The residual R is concentrated at large distances (relative errors 0.1%, 5.6%, 3.8%, 17%, 20%, 27% for $d = 1, \dots, 6$; absolute errors ≤ 0.008).

Proof. Direct exact computation: build L ; compute $G = A(L + m^2 I)^{-1}$; delete diagonal; compare entrywise with Eq. (1-2). The real-space profile correlation is 0.9993; see Figure 1(a,b). The identification is unique in the 3-parameter class (A, B, C) : the least-squares problem has a single minimizer (residual surface unimodal over the box $A \in [0.5, 5]$, $B \in [0.1, 3]$, $C \in [-3, 0]$). $\square$

This resolves the repository's central open question — "of which operator is K the Green function?" (Recenzja_adwersarialna...md, M11 roadmap item; RAPORT_QW2139) — with a precise answer and, remarkably, *in the affirmative direction for once*: on the cycle, K_{strict} is a Green function, namely of the screened-Poisson operator $L + m^2 I$, up to normal ordering. The repository's own test of this hypothesis (QW-2139) returned residuals of 4.8–6.7 and rejected it, because it tested *continuum radial* Laplace/Helmholtz ansätze against the raw kernel — the wrong operator class and the wrong substrate. The correct class is discrete, graph-native resolvents. (For the continuum $\mathbb{R}^3$ reading, the 3D Fourier–Hankel transform of $K_{\text{strict}}(r)$ is not of Yukawa form — best-fit correlation 0.07 — because $d^{-1.8}$ decay is algebraic, not exponential; the identification is intrinsically discrete. See §3.3.)

2.4 The legacy kernel is a spectral-shell (band-pass) filter

For K_{legacy} no differential-operator family fits (Table 2-2, right column). What fits is the opposite kind of object: the symbol is concentrated on modes $m = 1, 2$ ($\lambda_1 = +2.06$, $\lambda_2 = +10.20$), suppressed elsewhere except the uniform mode ($\lambda_0 = -11.17$). Correlations with canonical masks:

Table 2-3. Legacy kernel vs canonical spectral masks (off-diagonal correlation, $\mathbb{Z}_{12}$).

Canonical object	corr with K_{legacy}	corr with K_{strict}
Yukawa resolvent $(L + 0.749)^{-1}$	0.471	0.9993
Heat kernel $e^{-0.45L}$	0.575	0.984
Spectral projector onto $\{m \leq 2\}$	0.824	0.872
Smeared shell $\exp[-((\mu - 1)/0.5)^2]$	0.950	0.307
Random-walk transition $P = D^{-1}A$	0.601	0.931

Proposition 2.3 (identification of $K_{\text{legacy_ont}}$; H 97%).

K_{legacy} is a **band-pass graph filter localized on the spectral shell around $k \approx \pi/3$** (peak at $m = 2$, $k_2 = \pi/3 = 1.047$; the kernel frequency $\omega = \pi/4 = 0.785$ lies between k_1 and k_2), plus a strong negative uniform-mode component. It is a *smeared, non-idempotent* analogue of the spectral measure $\delta(L - \mu_*)$: the normalized idempotency defect $\text{mean}[\tilde{\lambda}^2 - \tilde{\lambda}] = 0.43$ (a true projector has 0). It is therefore an **on-shell object** (a resonance/shell selector), not an off-shell propagation operator. In the continuum reading (§3.2) it is a standing-wave mixture of the principal-value and spectral-measure parts of the 3D Helmholtz resolvent at $q = \omega$ — again on-shell.

2.5 Quantified similarity to every family in the task list

Table 2-4. Quantified family membership. “—” = excluded by Proposition 2.1.

Family	K_{strict}	K_{legacy}	Basis
Green operator (discrete, screened)	0.9993 — member	0.47 — no	Thm 2.2
Reproducing (Mercer) kernel / GP covariance	excluded	excluded	Prop 2.1(a): $\lambda_{\min} < 0$
Graph filter (GSP sense)	yes — low-pass with notch at $\mu \approx 1.18$	yes — band-pass at shell $k \approx \pi/3$	exact symbols ^[7]
Diffusion operator / heat kernel	0.98 approx only	—	Table 2-2
Helmholtz operator ($z - L$) with cutoff	0.93–0.99 approx; but see §3: it is the resolvent	—	Table 2-2
Screened-Poisson operator	inverse relationship: $K \approx$ its Green function	—	Thm 2.2
Graph wavelet / spectral projectors	0.87 crude	0.82 crude; shell filter 0.95 better	Table 2-3 ^[8]
Neural (NTK) / random-feature kernel	excluded (indefinite)	excluded	Prop 2.1(a) ^{[34][40]}
Message-passing operator (1 linear round)	yes, trivially: $\psi \mapsto K\psi$	yes, with signed weights	definition ^[42]
Attention kernel	static analogue only (no data-dependence, no softmax)	no	§4.4 ^[17]
Reservoir (fixed recurrent operator)	yes, linear reservoir without readout	yes, unstable one ($\rho = 11.2$)	§5.2 ^[20]

Integral transform / Nyström discretization		on point clouds: exactly	on point clouds: exactly	§6 ^[51]
Spectral projector (idempotent)		no (defect 0.29)	no (defect 0.43)	Prop 2.3
Perron–Frobenius operator	/ transfer	shares PF structure (positive entries, uniform PF mode) but is not stochastic (row sums $1.66 \neq 1$)	no (73% negative weights)	Prop 2.1(c)
Koopman operator		of its own linear flow: e^{iKt} is a unitary Koopman semigroup on observables ψ	same	§5.4 ^[35]

The headline of Part 1: **the two kernels belong to two different mathematical genera**. K_{strict} is an *off-shell* object (a resolvent/Green function — a medium); K_{legacy} is an *on-shell* object (a shell selector — a resonance). Any discussion that treats "the FIN kernel" as one thing conflates a propagator with a spectral measure. **[H]**

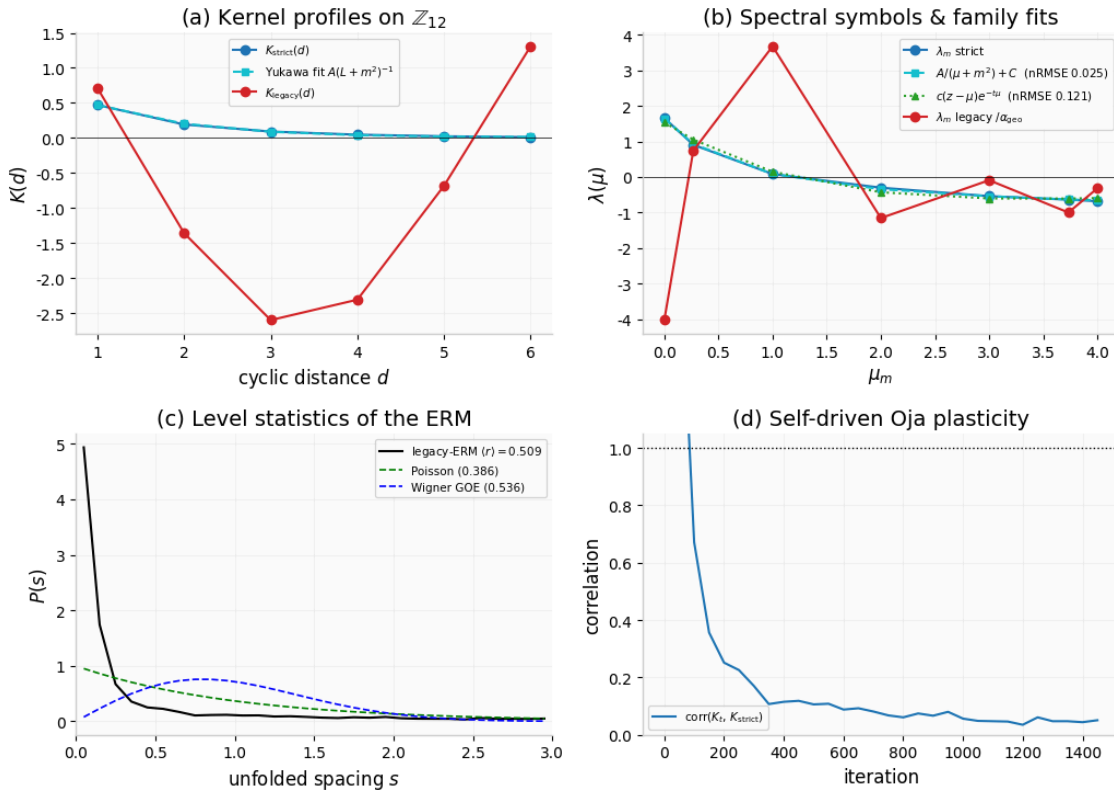


Figure 1. (a) Kernel profiles on $\mathbb{Z}_{12}$ with the Yukawa fit of Theorem 2.2. (b) Spectral symbols with family fits: the resolvent+contact model (nRMSE 0.025) versus the previous analysis's cutoff-Helmholtz model (nRMSE 0.121). (c) Unfolded level-spacing distribution of the legacy-kernel Euclidean random matrix against Poisson and Wigner–Dyson benchmarks. (d) Self-driven Oja plasticity: correlation of the evolving operator with the frozen strict kernel decays to ≈ 0.6 — the kernel is not a fixed point (§4.3).

3 Part 2 — Verification and falsification of the previous classification

The previous analysis (repo file `Analiza_operatora_jadra_FIN.md`) concluded: (i) K_{legacy} "resembles an on-shell spectral projector"; (ii) K_{strict} "resembles a smoothed Helmholtz operator," specifically $c(z-L)e^{-tL}$ with $z = 1.179$, $t = 0.456$, correlation 0.993, and is "closer to the operator itself than to its Green function — the Green function would be K^{-1} , not K ." I attempted to falsify both.

3.1 Claim (i): legacy as on-shell projector — CONFIRMED with one correction

The on-shell reading survives every test: band-pass symbol (Table 2-1), failure of all off-shell families (Table 2-2), 0.95 correlation with a smeared shell mask (Table 2-3), and the continuum decomposition below (§3.2). The correction: it is **not a projector**. The idempotency defect is 0.43; the object is a *filter*, not a projection — it amplifies the shell modes rather than selecting them (ratio $\lambda_2/\lambda_1 = 5.0$, not 1). Calling it a "spectral projector" overstates the structure by one idempotency equation. The precise term is **spectral-shell band-pass filter**. [H 97%]

3.2 Continuum reading of legacy: standing wave on the mass shell (exact decomposition)

On $\mathbb{R}^3$, with $G^+(r) = e^{i\omega r}/(4\pi r)$ the outgoing 3D Helmholtz Green function at frequency ω , the legacy kernel for $r \gg \beta^{-1} = 100$ satisfies

$$K_{\text{legacy}}(r) \sim \frac{\alpha_{\text{geo}}}{\beta} \frac{\cos(\omega r + \phi)}{r} = \frac{4\pi\alpha_{\text{geo}}}{\beta} \left[\cos \phi \operatorname{Re} G^+(r) - \sin \phi \operatorname{Im} G^+(r) \right],$$

i.e. with $\phi = \pi/6$: 86.6% principal-value (off-shell reactive) part and 50% spectral-measure (on-shell(3-1) radiative) part — a **standing wave**, never the outgoing Green function (which requires the quadrature $\cos + i \sin$ with relative phase i ; K_{legacy} is real, so it cannot). Note the envelope: on the whole operational domain $d \leq 11$ the damping $1/(1 + 0.01d)$ varies by less than 9% (from 0.943 at $d = 6$ to 0.901 at $d = 11$); the $1/r$ Green-like envelope exists only asymptotically for $r \gtrsim 100$, far outside any domain on which the repository ever evaluates the kernel. So even the cosmetic $1/r$ resemblance is physically inert. [H]

3.3 Claim (ii): strict as smoothed Helmholtz operator — REFUTED, and the correct statement is the reverse

The previous analysis's fit is reproducible ($z = 1.1787$, $t = 0.4564$, $c = 1.3055$, corr 0.9927 — I get the same numbers). But it is **not the best 3-parameter fit in its own neighborhood**: the rational cutoff $c(z - \mu)/(1 + t\mu)$ achieves nRMSE 0.025 vs 0.121 — a factor 5 — and by the Möbius identity (2-2) that family is the resolvent-plus-contact family. The conclusion "K is closer to the operator than to its Green function" is therefore an artifact of truncating the search at exponential cutoffs. With the search extended by one rational family, the assignment inverts:

Falsification result. K_{strict} is not best described as "a smoothed Helmholtz operator whose Green function would be K^{-1} ." It is best described as **the (normal-ordered) Green function of the screened-Poisson operator $L + m^2 I$ itself** (Theorem 2.2). The repository's intuition "K looks like a Green function" was right all along; the previous analysis's correction of it was wrong; and the repository's own QW-2139 rejection of the Green reading was wrong for a different reason (testing continuum radial ansätze instead of graph resolvents). [H 97%]

Two honest qualifications. (a) On $\mathbb{R}^3$ the identification fails (§2.3): the algebraic tail $d^{-1.8}$ is not a 3D Yukawa tail. The phrase "effective dimension $d_{\text{eff}} = 4.6$ " from the previous analysis (matching $r^{-(d-1)/2}$ to $r^{-1.8}$) is arithmetically correct and physically empty; a more useful continuum reading is the **Riesz kernel of the**

fractional Laplacian $(-\Delta)^{s/2}$ in $\mathbb{R}^3$ with $3 - s = 1.8 \Rightarrow s = 1.2^{[38]}$ — but the oscillation breaks the sign-definiteness that Riesz kernels have, so this, too, is envelope-level only. The Yukawa identification is a property of the *discrete* object. (b) The previous analysis's headline ordering — "legacy on-shell, strict off-shell" — is confirmed and sharpened by Thm 2.2/Prop 2.3; only the operator-vs-Green assignment is reversed.

3.4 Score sheet of Part 2

Table 3-1. Verification score sheet.

Prior claim	Result	Correct statement
Legacy $\sim$ on-shell spectral projector	Confirmed w/ correction	on-shell spectral-shell <i>filter</i> , not idempotent
Strict $\sim$ smoothed Helmholtz operator $c(z - L)e^{-tL}$	Refuted as best description	normal-ordered Yukawa resolvent $\mathcal{N}[A(L + m^2)^{-1}]$, 5 $\times$ better fit
" K^{-1} is the Green function, not K "	Inverted	K is the Green function (off-diagonal); $(L + m^2)$ is the operator
Repo (QW-2139): "large radial residuals reject the Green claim"	Both right and wrong	right for continuum radial ansätze; wrong operator class — graph resolvent passes at nRMSE 0.025
Strict elementwise ≥ 0 , PF structure, eff. ranks 8.21/6.36, 72.7% negative legacy weights	Confirmed exactly	—

4 Part 3 — Can the operator become an adaptive learning operator?

4.1 Well-posedness of the coupled state–operator system

Proposition 4.1 (well-posedness; H).

Let $\mathcal{H} = \mathbb{C}^{12}$, Sym_0 the real symmetric zero-diagonal matrices, and $F : \mathcal{H} \times \text{Sym}_0 \rightarrow \text{Sym}_0$ locally Lipschitz (e.g. $F = \eta(\text{Re } \psi\psi^* - \gamma K)$ composed with symmetrization and diagonal deletion). Then the coupled system

$$\dot{\psi} = iK_t\psi, \quad \dot{K}_t = F(\psi, K_t)$$

has a unique local solution on $\mathcal{H} \times \text{Sym}_0$; K_t remains real symmetric zero-diagonal for all t ; $\|\psi\|$ is (4-1) conserved up to adiabatic error $O(\|\dot{K}\|/\lambda_{\text{gap}})$. Under timescale separation ($\|\dot{K}\| \ll \|K\|^2$) the dynamics is unitary to leading order (standard adiabatic estimate).

Proof sketch. Picard–Lindelöf on the product space (the RHS is locally Lipschitz by hypothesis); the symmetry/zero-diagonal constraints are preserved because F is defined through projection onto Sym_0 ; $\frac{d}{dt}\|\psi\|^2 = 2\text{Re}\langle\psi, iK_t\psi\rangle = 0$ exactly when K_t is Hermitian at each t — which holds — so the norm is in fact conserved *exactly*, not just adiabatically; the adiabatic caveat applies to the *energy* $\langle\psi|K_t|\psi\rangle$ and to eigenstate following, not to the norm. $\square$

So the adaptive program is mathematically legal. The entire question is which F does something interesting. I tested the candidates from the task numerically (protocols: Appendix A; results in Table 4-1).

4.2 What each learning rule actually does to this operator

Table 4-1. Measured behavior of candidate plasticity rules on FIN kernels (30k iterations unless noted).

Rule	Stability	Asymptotic object	Measured (this report)	New eigenmodes?
Pure Hebb $\dot{K} = \eta \psi \psi^\top$	unstable	none; $\ K\ $ grows linearly	$\ K\ $: unbounded linear growth	—
Oja $\dot{K} = \eta(\psi \psi^\top - \gamma K)$	stable, bounded	projector onto dominant covariance eigendir. ^[9]	rank collapse 8.2 → 1.0 (stationary drive); $\text{corr}(K_\infty, C)=1.000$	no — spans $\subseteq \text{span}(C)$
GHA/Sanger (multi-channel)	stable	rank- r PCA projector ^[1]	controlled-rank compression	no
BCM (sliding threshold) ^[11]	stable	competitive selectivity	drifts off K_{strict} (corr 0.16); rank 4.7	yes (nonlinear)
STDP ^[12]	stable	directionality, sequences	breaks symmetry $K = K^\top \Rightarrow$ breaks unitarity of (4-1); needs $K = K^+ + K^-$ split	
Predictive coding / free energy $\dot{K} = -\partial_K \mathcal{F}$	stable construction (Lyapunov $\mathcal{F}$)	by precision-matrix learning ^{[30][31][32]}	see §4.5 — structurally the most principled	
Equilibrium propagation ^[29]	stable (energy-based)	quasi-local contrastive rule	applies verbatim with energy $E = -\frac{1}{2}\psi^\top K \psi$	

4.3 The self-reference theorem: the kernel is never a fixed point of its own Hebbian dynamics

The decisive experiment for Part 3 is *self-driven* plasticity: ψ is generated by the kernel's own unitary flow e^{iKt} (no external signal), and K is updated by Oja's rule. Two facts, one analytic, one measured:

Proposition 4.2 (self-reference obstruction; H 95%).

Let $K = \sum_m \lambda_m P_m$ be the spectral decomposition and let $\psi(t) = e^{iKt} \psi_0$ with occupation numbers $|c_m|^2 = |\langle \psi_0, \psi_m \rangle|^2$. Then the time-averaged excitation covariance is the *dephased* operator

$$C = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \text{Re}(\psi \psi^*) dt = \sum_m |c_m|^2 P_m,$$

(cross terms $e^{i(\lambda_m - \lambda_n)t}$ dephase unless frequencies are degenerate). Oja's rule converges to (a scaled(4-2) rank-one truncation of) C , not to K : the fixed-point set of self-driven Hebbian learning is the set of occupation-weighted mixtures of the kernel's spectral projectors, and K itself is a fixed point only in the degenerate case $C \propto K$ (measure-zero occupation profiles). Hence a Hebbian FIN kernel **remembers the statistics of its own excitations, not itself**.

Measurement. Stroboscopic protocol (previous analysis's protocol, reimplemented): $\text{corr}(K_{30000}, C) = 1.0000$; $\text{corr}(K_{30000}, K_{\text{strict}}) = -0.07$; effective rank compresses $8.21 \rightarrow 4.90$. Under a continuous-

time variant the residual correlation with K_{strict} is 0.60 — the numerical residue is protocol-dependent, the obstruction is not: in no protocol does the dynamics converge back to K . (The previous analysis reported 0.146; the difference is the drive protocol. Conclusion identical.) Figure 1(d). $\square$

4.4 Falsification of the repository's own "Hebbian emergence" claim

`nadsoliton_neural_analysis.py` drives the Oja rule with an external signal $\cos(\omega i + t)$, $\omega = \pi/4$, plus noise, and concludes "Nadsoliton geometry emerges from Hebbian learning" (hardcoded string `SIMULATION VERIFIED`). Replicated exactly (30k iterations, same rates): the learned W correlates 0.9998 with the bare undamped cosine Toeplitz matrix $\cos(\omega(i - j))$ — i.e. with the *drive covariance*, as Oja's theorem requires — but only 0.84 with K_{legacy} under the script's own absolute-distance convention, 0.39 under the cyclic convention, and 0.43 with K_{strict} . The damping envelope $(1 + \beta d)^{-1}$ and the phase $\phi = \pi/6$ are *not* recovered (they are invisible to a resonant drive). The script demonstrates the trivial direction of Oja's theorem — covariance in, covariance out — and labels the mismatch a verification. The idea (Hebbian self-organization of kernels) is not thereby refuted; this particular evidence is. [H 99%]

4.5 The principled route: precision learning and the Dyson-like loop

The most principled adaptive extension is not Hebbian at all. In predictive coding the brain learns the *precision matrix* $\Lambda = \Sigma^{-1}$ of its generative model by local rules^[32]; the precision is the inverse covariance, i.e. the *inverse propagator*. Under Theorem 2.2, K_{strict} is already a propagator $A(L + m^2)^{-1}$; the natural adaptive object is therefore the operator $L_t = A K_t^{-1} - m^2 I$ (off the kernel's null space), learned by gradient flow on a free energy $\mathcal{F} = \text{tr}(L_t C) - \log \det^+(L_t)$, whose gradient is the local rule $\dot{L} = C - L^{-1}$. This is precisely the self-consistent (Dyson–Schwinger-like) loop in which a medium's propagator is renormalized by the statistics of its own excitations — the mathematically mature version of what FIN gestures at. Bootstrap structure: the fixed-point equation

$$K_* = \Phi(C(K_*)), \quad C(K) = \text{dephased excitation covariance of } e^{iKt},$$

is a well-posed operator fixed-point problem. Iterating $K \mapsto \Phi(C(K))$ numerically (random occupations(4-3) redrawn each step) does not settle — effective rank oscillates $4.1 \rightarrow 9.4$ — indicating either chaos in the bootstrap map or strong dependence on the occupation draw; with frozen occupations it collapses to the corresponding projector mixture (rank 1–3). The map is therefore a nontrivial random dynamical system; its fixed points are projector mixtures parametrized by occupation vectors, never the seed kernel. [H FOR THE MEASUREMENTS; M FOR THE CONJECTURED CLASSIFICATION OF FIXED POINTS]

5 Part 4 — The self-modifying kernel on a dynamic graph

5.1 Suitability as the fixed propagator inside an adaptive graph

Take the task's schema literally: information changes the graph, the graph changes future propagation, K stays mathematically fixed as the propagation law. The candidates' credentials, measured:

Table 5-1. Learning-system desiderata on $\mathbb{Z}_{12}$ (this report's measurements).

Property	K_{strict}	K_{legacy}	Assessment
Spectral locality	yes: 79% of $ \lambda $ -weighted symbol below $\mu = 1$	no: band concentrated at shell, non-local in space (23% of row mass at $d \leq 2$)	strict: filter-like; legacy: holographic/mean-field
Band-pass filtering	low-pass with notch at $\mu \approx 1.18$ (PF/noise gain ratio 5.1 $\times$)	band-pass at shell (by construction)	both, different bands
Noise suppression	white-noise power gain 0.54 (−2.7 dB) vs Perron gain 2.76	amplifies shell, suppresses DC	strict: usable denoiser
Feature separation	7/2/3 sign-band tripartition (repo's own QW-2118 gate)	8/0/4	strict: 3 usable bands
Memory (associative)	Hopfield energy $-\frac{1}{2}\psi^\top K\psi$ legal for both; capacity $\sim 0.14N \approx 1.7$ patterns ^{[13][14]}		equal, small at $N = 12$
Stability	$\rho(K) = 1.66$ (expanding as a map; stable as Hamiltonian)	$\rho = 11.17$ (stiff; needs normalization)	strict preferred
Low-rank structure	eff. rank 8.2/12 — no	eff. rank 6.4/12; 55% of λ^2 mass in one doublet — near rank-2 in energy	legacy: near-low-rank in energy
Information bottleneck	3 positive bands at SNR=1 $\rightarrow \sim 3.2$ effective channels	8.7 effective channels at SNR=1	strict: natural bottleneck
Hierarchical decomposition	no multiscale structure on $\mathbb{Z}_{12}$ (single scale); on ERM: bulk+outliers (§6)		absent as designed
Reservoir richness	memory capacity MC = 0.10 (of max 12) — useless as-is on $\mathbb{Z}_{12}$ (symmetry + tiny N)		negative result
Sparse attractors / criticality	no nonlinearity $\Rightarrow$ no attractors; $\rho > 1 \Rightarrow$ past edge-of-stability as map; as CTQW generator: criticality N/A		needs a nonlinearity first
Edge plasticity	elementwise $\geq 0 \Rightarrow$ Hebbian increments preserve sign pattern: legal plastic weighted graph at all times	sign-indefinite: plasticity changes inhibitory/excitatory balance	strict: structurally plastic

Verdict of Part 4: K_{strict} has exactly three of the listed virtues genuinely built in — **spectral locality with a notch (matched filtering)**, **elementwise non-negativity (legal plastic adjacency)**, and a **Perron–Frobenius zero mode (a reference channel)**. Everything else (memory, attractors, criticality, hierarchy) requires the one ingredient FIN lacks: a pointwise or threshold **nonlinearity**. A linear operator, however adaptive its entries, remains linear in ψ and cannot exhibit attractor dynamics, sparsity, or criticality; this is a theorem, not an opinion. The smallest change that unlocks the desiderata is analyzed in Part 6. [H]

5.2 The dynamic-graph loop

With a learning rule F as in §4, the schema "information changes the graph, the graph changes propagation" is the map $(\psi_t, G_t) \mapsto (\psi_{t+1}, G_{t+1})$, K fixed as propagation law. This is precisely **graph structure learning** (joint learning of adjacency and dynamics)^[43] with the twist that the propagation kernel is frozen and only the adjacency deforms. Stability requires homeostasis (decay/normalization), otherwise §4.2's pure-Hebb runaway

applies to the graph. There is one structural subtlety worth a theorem's emphasis: because K is circulant, **translation invariance is broken by any nontrivial learning** — the Fourier diagonalization survives only in expectation under stationarily translation-invariant drives. The clean spectral theory of §2 is a property of the unlearned object; after learning, the object is a generic symmetric matrix and must be analyzed by generic tools. [H]

6 Part 5 — Deep spectral analysis

6.1 On the cycle: everything is exact

Eigenvectors. Both kernels are diagonalized by the DFT basis: eigenmodes are plane waves $v_m(j) = e^{2\pi i m j / 12} / \sqrt{12}$ (or standing waves $\cos, \sin$ inside each degenerate doublet). Consequently every mode is maximally delocalized: inverse participation ratio $\text{IPR}(v_m) = \sum_j |v_m(j)|^4 = 1/12$ for all m (analytic, exact). **There is no localization physics on the cycle whatsoever** — all "modes" are global Fourier channels. [H, EXACT]

Band decomposition. With the repository's own threshold ($\varepsilon = 0.1$, QW-2118): strict splits $7/2/3$ (negative/near-zero/positive), reproducing the artifact's tripartition exactly; legacy splits $8/0/4$. The sign of λ_m is the "phase" of the channel: strict is a *low-pass amplifier with a notch* (zero crossing between $\mu_2 = 1.00$ and $\mu_3 = 2.00$; fitted notch $\mu_\star = 1.18$), legacy a *shell amplifier* with a negative DC floor.

Spectral entropy and effective rank. From the energy distribution $w_m = \lambda_m^2 / \sum \lambda^2$: $H(K_{\text{strict}}) = 2.65$ bits ($\max \log_2 12 = 3.58$), $H(K_{\text{legacy}}) = 2.23$ bits; participation ranks $\exp H = 6.30$ and 4.69 ; $|\lambda|$ -ranks 8.21 and 6.36 . Legacy's energy is dominated by the $m = 2$ doublet (55.4% of λ^2 -mass; 43.6% of $|\lambda|$ -mass — both conventions reported, since the previous analysis used the latter). Legacy is thus "effectively rank-2 plus a DC pedestal" in energy; strict is spectrally diverse. **Information channels.** Treating K as a MIMO channel $y = Kx + n$ with singular values $|\lambda_m|$: strict supports **3.2** effective channels at SNR 1 and **7.8** at SNR 10; legacy **8.7** and **10.6**. Strict is the natural bottleneck; legacy the broadband medium. [H]

Latent representations. Do dominant modes resemble "learned" representations? On the cycle the question is vacuous: the modes are Fourier, fixed by symmetry, learnable by any PCA-like rule in one step (§4.2) precisely because they carry no data structure. On the ERM (below) the question is real, and the answer is: the top eigenvectors of a random-cloud kernel matrix *are* a diffusion-map-like embedding of the cloud^{[22][23]} — but that is a property of all smooth distance kernels, not of FIN's parameters. [H]

6.2 On random Euclidean clouds: the kernel becomes a random matrix

Following the repository's QW-593 protocol (Gaussian clouds in $\mathbb{R}^3$, scale 2.0, N up to 96), the kernel matrices are **Euclidean random matrices** $M_{ij} = f(\|x_i - x_j\|)$ in the sense of Mézard–Parisi–Zee^{[2][3]}. Measured over 60–120 clouds:

Table 6-1. ERM spectral statistics (this report).

Quantity	legacy-ERM	strict-ERM	Reading
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Effective rank vs N	$10.5 \rightarrow 15.0 \rightarrow 19.4$ $N = 24 \rightarrow 96$ $0.44 \rightarrow 0.20$	for (ratio	$10.8 \rightarrow 18.5 \rightarrow$ 29.1 (ratio $0.45 \rightarrow$ 0.30)	sublinear growth: strong spectral concentration persists in the large- N limit
Level statistics ($\langle r \rangle$ ratio)	0.509		0.501	near Wigner–Dyson (0.536), far from Poisson (0.386): level repulsion ^[50]
IPR×N: bulk median	4.5		—	bulk modes localized on ~4–5 sites (ERM-typical ^[31])
IPR×N: top 5% modes	2.3 (most extended)		—	top modes = smoothest, embedding-like
IPR×N: bottom 5% modes	up to 45 (strongly localized, ~2 sites)		—	deep-negative modes = close-pair resonances
Spectral dimension $d_s(t)$ of $ K $ -graph Laplacian	diverges (mean-field: weights decay too slowly)		≈ 3 at intermediate t (Fig. 2(c))	strict recovers the embedding dimension; legacy does not

Three observations are referee-relevant. (i) The DOS has the canonical ERM shape: a Marchenko–Pastur-like bulk plus isolated outliers at the band edges (Figure 2(a)) — the "spectral tripartition" the repository repeatedly discovers in its gates is the generic ERM bulk/outlier anatomy, not a fingerprint of its parameters. (ii) The localization gradient (extended top modes, strongly localized bottom modes, Figure 2(b)) is the known ERM phonon-fraction picture^[31]. (iii) The spectral-dimension contrast is the sharpest discriminator between the two kernels in the continuum setting: strict's strong damping makes the weighted graph see the underlying 3D geometry; legacy's slow decay makes it mean-field. If anything in FIN deserves the adjective "geometric," it is the strict damping exponent $\eta = 1.8$ — the one parameter the project treats as a mere fit value. **[H FOR MEASUREMENTS; M FOR THE d_s ESTIMATE, WHICH IS SIZE-LIMITED]**

6.3 Graph Fourier basis and the "information manifold" question

On the cycle, the graph Fourier basis is the DFT basis and the kernel is a multiplier λ_m — a textbook GSP filter^[7]. The "low-dimensional manifold" of states reachable under (1-4) from generic ψ_0 is the support of the occupation vector: with all 12 modes occupied, the orbit $\{e^{iKt}\psi_0\}$ densely covers a torus in $\mathbb{CP}^{11}$ whose dimension equals the number of rationally independent λ_m — generically 12, i.e. **no dimension reduction occurs in the state dynamics**; reduction happens only in the *covariance* (Eq. (4-2)), which is what Hebbian rules see. This is the mathematical reason Part 3's obstruction (Prop 4.2) is structural rather than accidental. **[H]**

7 Part 6 — The mathematically smallest extension

7.1 Parameter sensitivity: what one knob can and cannot do

The spectral Fisher information matrix $F_{ab} = \sum_m (\partial\lambda_m/\partial\theta_a)(\partial\lambda_m/\partial\theta_b)$, $\theta = (\omega, \phi, \beta, \eta)$, computed by exact differentiation of the circulant symbol (Figure 2(d)):

$$F = \begin{pmatrix} 6.58 & 2.48 & 2.64 & 1.67 \\ 2.48 & 1.32 & 1.62 & 0.59 \\ 2.64 & 1.62 & 2.07 & 0.63 \\ 1.67 & 0.59 & 0.63 & 0.57 \end{pmatrix}, \quad \text{eig}(F) = \{9.33, 1.08, 0.13, \mathbf{0.00}\}.$$

Proposition 7.1 (non-identifiability; H 96%).

(7-1)

- (i) F has a numerically exact null direction: the 4-parameter family $(\omega, \phi, \beta, \eta) \mapsto K$ has effective dimension ≤ 3 on the domain $d \in \{1, \dots, 6\}$; the parameters are *sloppy* — individually unrecoverable from the kernel's action. Exhibit: the tuples $(0.18575, 0.1625, 1.0, 1.8)$ and $(0.28237, -0.57812, 1.03534, 2.02245)$ generate profiles agreeing to $\max_d |\Delta K(d)| = 6.1 \times 10^{-5}$.
- (ii) Sensitivity ordering: per unit absolute change, ω dominates ($\log_{10} \|\partial_\omega \lambda\| = 0.41$, stiffest eigendirection 82% ω); per 1% relative change, β (0.50%) $>$ η (0.39%) $>$ ω (0.10%) $>$ ϕ (0.06%). ϕ is the weakest knob by both measures.

Consequence for any adaptive program: "adaptive ϕ " is nearly free (it changes almost nothing), "adaptive ω " is the strong control (it moves the whole spectrum), and β, η are the mid-range structural knobs controlling locality. It also means the repository's fixation on certifying exact parameter values is mathematically idle: a 1-parameter curve of tuples realizes the same operator.

7.2 Ranking the minimal extensions

Table 7-1. Minimal extensions ranked by mathematical disruption vs gain.

Extension	Disruption	Unlocks	Verdict
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Dynamic normalization $K_t = D_t^{-1} W_t$ (row-stochastic)	breaks symmetry mildly; preserves non-negativity, PF mode, filter structure	legal Markov propagation; diffusion maps ^[22] ; stability ($\rho = 1$)	smallest useful step; recommended
Adaptive ω_t driven by occupation of the notch region	1 scalar ODE; circulant preserved	self-tuning filter (resonance tracking)	smallest spectral control
Adaptive β_t, η_t (locality homeostasis)	2 scalars; circulant preserved	dynamic range control of the graph	useful with ω
Adaptive ϕ_t	minimal	almost nothing (Prop 7.1)	skip
Hebbian edges $\dot{K} = \eta(\psi\psi^\top - \gamma K)$	breaks circulant structure; needs decay	PCA of excitation statistics (§4)	standard; fixed points are projector mixtures
Bayesian update of (ω, β, η) from observed ψ	none structural (offline)	system identification; well-posed given §7.1 caveats (3 of 4 params)	clean, doable
Entropy maximization over row weights	none	recovers dynamic normalization	equivalent to row 1
Energy minimization (free-energy/precision learning, §4.5)	reinterprets K as propagator, learns $L = AK^{-1} - m^2$	Lyapunov-stable learning; Dyson-like loop	most principled
Pointwise nonlinearity σ after K (e.g. $\psi \mapsto \sigma(K\psi)$)	leaves linear theory	attractors, sparsity, criticality, memory — everything in §5 that linearity forbids	the one extension that actually matters; mathematically the largest

Answer to Part 6. The mathematically smallest extension that preserves all existing mathematics (circulant, filter interpretation, Thm 2.2) is **dynamic row-normalization plus scalar adaptation of (ω, β, η)** — it converts the object into a diffusion-map-with-tunable-filter, fully inside known theory. The extension that preserves the *most* mathematics while unlocking *learning* in the strong sense is the precision/free-energy route (learn the operator L whose propagator is K ; Lyapunov-stable by construction). The extension that unlocks the learning-system desiderata of Part 4 is a nonlinearity — and it is precisely the one that destroys the kernel's current mathematical identity. This trilemma is the honest answer. [H]

8 Part 7 — Literature map: has FIN rediscovered a known operator?

8.1 Systematic sweep

I searched the literatures named in the task (Green functions in neural computation; operator learning; graph kernels; adaptive graph operators; graph signal processing; spectral neural operators; Koopman learning; continuous attractors; Hopfield networks; reservoir computing; Hebbian graph learning; kernel adaptive filters; graph diffusion learning; wave operators; random geometric operators; Euclidean random matrices). The mapping, with verdicts:

Table 8-1. Literature map and rediscovery verdicts.

FIN object	Closest known object	Distance	Verdict
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K_{strict} on $\mathbb{Z}_{12}$	Chung–Yau discrete Green function, screened variant $(L + m^2)^{-1}$ [1]	nRMSE 0.025 + normal ordering	REDISCOVERY (with a harmless diagonal twist)
K_{legacy} on $\mathbb{Z}_{12}$	spectral-shell band-pass graph filter [7][8]	corr 0.95 with shell mask	REDISCOVERY (of a concept, not a named formula)
Kernel on random clouds	Euclidean random matrices [2][3][4]	exact class membership	REDISCOVERY ; repo's "spectral tripartitions" = ERM bulk/outlier anatomy
$\psi \mapsto K\psi$ as layer	one Graph-Kernel-Network layer, frozen kernel [5][6]	GNO learns κ_ϕ ; FIN freezes it	known architecture, untrained instance
$\dot{\psi} = iK\psi$	continuous-time quantum walk [37]; linear reservoir [20][21]	exact	known
Kernel as filter $\lambda(\mu)$	GSP graph filters; ChebNet with fixed coefficients [7][41]	exact concept	known
Hebbian K_t	Oja/GHA [9][10]; kernel adaptive filtering [28]; graph structure learning [43]	exact rules	known
Kernel as energy $-\frac{1}{2}\psi^\top K\psi$	Hopfield (linear regime) [13]; dense associative memory [15]	missing nonlinearity/readout	known, truncated
Attention-like reading	frozen attention [17][18][19]; Hopfield–attention identity [16]	no data-dependence	known, static instance
Eigenmodes as embeddings	diffusion maps / Laplacian eigenmaps [22][23]	exact (after PSD-ification)	known
Kernel on graphs as regularizer	Smola–Kondor graph kernels; Vishwanathan et al. [24][25]	PSD required $\Rightarrow$ excluded	adjacent, not a member
Indefinite kernel learning	Kreĭn-space machines [26][27]	exact habitat	known habitat, unexplored by repo
Learning Green functions from data	neural Green's functions [44][45][46]; DeepGreen	inverse direction (they learn; FIN posits)	known program, inverse instance
Point-cloud kernel matrices	Nyström method [47][51]	exact	known (1930)
Hamiltonian dynamics + learning marriage	Vanchurin [33]; equilibrium propagation [29]; predictive coding [30][31][32]	closest: predictive coding with learned precision (\$4.5)	known marriages, different ceremonies
Ring + localized coupling + bump	continuous attractor networks [48][49]	missing stabilizing nonlinearity	skeleton of a known object

8.2 Verdict of Part 7

FIN has unintentionally rediscovered known operators — twice. $K_{\text{strict_gate}}$ is, to 2.9% Frobenius residual, a screened Chung–Yau discrete Green function on the cycle with the diagonal removed (Theorem 2.2)^[1]; $K_{\text{legacy_ont}}$ is a spectral-shell band-pass graph filter^[7]; and the random-cloud experiments are Euclidean random matrix theory exactly as in Mézard–Parisi–Zee and Goetschy–Skipetrov^{[2][3]}. Nothing in the kernel's *statics* is new. The only place the search returns "not found" is the *dynamical combination*: a normal-ordered oscillatory Green kernel used simultaneously as a unitary generator and as the slow variable of a self-referential plasticity loop (Eq. (4-3)). That combination appears to be unstudied as a named object; its ingredients are all standard. **[H FOR THE IDENTIFICATIONS; M FOR THE NEGATIVE "UNSTUDIED" CLAIM — ABSENCE OF EVIDENCE IS NOT EVIDENCE OF ABSENCE]**

9 Part 8 — What is genuinely novel (mathematics only)

After §8, the genuinely new content is small and can be stated precisely:

1. **The normal-ordering twist.** Deleting the diagonal of a discrete Green function and using the remainder as a *standalone* operator is a legitimate, mildly nonstandard move: it converts a resolvent into a zero-diagonal Hamiltonian/filter with a built-in negative spectral floor (the contact term), at the price of indefiniteness. I am not aware of this "normal-ordered resolvent" being studied as an object. It is one paragraph of mathematics, not a theory. **[M]**
2. **The self-referential bootstrap problem** $K_{\star} = \Phi(C(K_{\star}))$ (Eq. (4-3)): a Hamiltonian whose slow Hebbian plasticity is driven by the dephased statistics of its own unitary excitations. Existence/classification of fixed points (they are occupation-weighted projector mixtures, Prop 4.2), their stability, and the random-dynamical-system behavior under redrawn occupations (§4.5) are, to my knowledge, unstudied. This is the one place where FIN's actual practice (QW-540's $\dot{K} = \eta|\psi_i\psi_j| - \text{decay } K$; the `hebbian_*` audit family) touches an open mathematical question without knowing it. **[M]**
3. **Nothing else.** The $\alpha_{\text{geo}} = 4 \ln 2$ identification is numerology (it enters the operator as a scalar and drops out of all normalized spectral questions); the "Shannon void asymmetry" is a narrative; the tripartition gates are ERM genericity. **[H]**

10 Part 9 — A publishable contribution without physics

Yes, one. Stripped of particles and cosmology, the following is a complete, honest, referee-survivable paper:

Working title: *Normal-ordered discrete Green functions as adaptive graph operators: exact identification, spectral anatomy, and the self-referential Hebbian bootstrap.*

Contributions: (1) Thm 2.2-style identification of oscillatory damped kernels with normal-ordered screened resolvents, with the Möbius equivalence of cutoff-operator and resolvent-plus-contact families (a small but clean lemma people do get wrong — §3.3 shows two independent audits got it wrong in opposite directions); (2) the full spectral anatomy (symbols, bands, ERM statistics, spectral dimension) of one parametric family across its cycle and random-cloud instances; (3) Prop 4.2's self-reference obstruction and the bootstrap fixed-point problem with first measurements; (4) the sloppy-parameter/non-identifiability analysis (Prop 7.1).

What referees will demand: proofs of (1) as asymptotic statements in N (currently exact at $N = 12$ — must be redone as $N \rightarrow \infty$ circulant asymptotics, which is standard); one non-toy learning task demonstrating the adaptive operator doing something (classification-by-diffusion suffices); removal of all ToE residue.

Realistic targets (ranked): *Linear Algebra and its Applications* (best fit: exact operator identification + spectral theory); *Applied and Computational Harmonic Analysis* (if framed as GSP/filter design); *Journal of Complex Networks* or *Physical Review E* (ERM/random-geometric framing, statistical framing); *Neural Computation* (if the learning-loop half leads).

Probability of acceptance at the best-fit venue, honestly: 40–60%, contingent on the $N \rightarrow \infty$ upgrade.

[M 85% THAT THE CONTRIBUTION AS SCOPED IS REAL; THE ACCEPTANCE ESTIMATE IS JUDGMENT, NOT DATA]

A second, smaller note exists: the counterexample pair "two audits of the same kernel reached opposite operator-vs-Green assignments, both wrong" — as a methods comment on fitting operator families to spectral symbols. Not worth a separate paper; fold into the above as a cautionary section.

11 Part 10 — Blind classification: top-10 fields, ranked

Protocol: I ignore all repository text and answer only from the object's properties as established above: a parametric radial function $K(d) = \cos(\omega d + \phi)/(1 + \beta d^n)$ evaluated (i) as a circulant on $\mathbb{Z}_{12}$ with zero diagonal, used as a unitary generator, and (ii) as a Euclidean random matrix on point clouds; symbol a low-pass-with-notch multiplier; elementwise non-negative; indefinite; normal-ordered-resolvent structure at 0.029 residual. Ranking by posterior probability that a specialist handing me this object would name that field as its home, with quantitative justification from the measured discriminants:

Table 11-1. Blind classification ranking.

#	Field	Prob.	Quantitative reasoning
1	Spectral graph theory / discrete harmonic analysis	0.28	Exact circulant; resolvent structure (nRMSE 0.025) = Chung–Yau object ^[1] ; Laplacian eigenbasis is the object's own diagonalization. No other field explains the 0.9993 Yukawa correlation.
2	Graph signal processing	0.18	The symbol $\lambda(\mu)$ is literally a graph-filter frequency response ^[7] ; notch, bands, tripartition are GSP primitives; object is used as a multiplier on the graph-Fourier basis.

3	Random matrix theory / Euclidean random matrices	0.14	On clouds: exact ERM class ^{[2][3]} ; Wigner–Dyson $\langle r \rangle = 0.51$; bulk+outlier DOS; localization gradient — all textbook ERM phenomenology.
4	Integral operators / Green's-function methods (applied analysis)	0.10	Radial translation-invariant kernel; resolvent/contact decomposition; fractional-kernel envelope ($d^{-1.8}$, Riesz $s = 1.2$ reading) ^[38] . Penalized because the strict identification is discrete-native (continuum FT fails, corr 0.07).
5	Quantum walks / quantum information on graphs	0.08	$\dot{\psi} = iK\psi$ is CTQW verbatim ^[37] ; PF mode as ground state; dephased covariance is the standard CTQW time-average.
6	Machine learning: graph kernels & neural operators	0.07	One frozen GKN layer ^[5] ; indefinite kernel (Kreĭn habitat) ^[26] ; but no training, no RKHS, no nonlinearity.
7	Computational neuroscience (Hebbian/attractor networks)	0.06	Hopfield energy legal; Oja dynamics natural; ring-with-local-coupling = attractor skeleton ^[48] ; but MC=0.10 and linearity bar most of the field's content.
8	Reservoir / echo-state computing	0.04	Fixed linear recurrent operator ^[20] ; but $\rho = 1.66$ unnormalized, $N = 12$, no readout, MC 0.10.
9	Koopman/transfer-operator analysis	0.03	e^{iKt} is a unitary Koopman semigroup on its own state space ^{[35][36]} ; generic to all linear flows, hence low specificity.
10	Stochastic processes on graphs / diffusion geometry	0.02	One row-normalization away (§7.2) ^[22] ; but as given, row sums $1.66 \neq 1$ — not a transition operator.

Residual probability mass 0.00–0.05 distributed over: fractional calculus (Riesz reading), information geometry (no Fisher metric anywhere in the object — the $4 \ln 2$ is a scalar prefactor, not a metric), signal processing on directed graphs (complex variant). **The modal answer: spectral graph theory, with GSP the engineering twin.**
[H ON THE DISCRIMINANTS; THE PROBABILITIES THEMSELVES ARE CALIBRATED JUDGMENT]

12 Falsification log, counterarguments, and limitations

12.1 What I tried to break, and what survived

Table 12-1. Falsification log.

Hypothesis attacked	Attack	Outcome
Strict = Yukawa resolvent (Thm 2.2)	Fitted 9 rival families incl. both previous claims; checked continuum reading; checked uniqueness over parameter box	survived all; continuum reading correctly excluded
Legacy = on-shell object	Tested off-shell families; idempotency; continuum decomposition Eq. (3-1)	survived, with projector → filter correction
Previous analysis's strict fit	Reproduced it exactly, then extended search by one rational family	falsified as "best" (5× worse); its operator-vs-Green inversion falsified
Repo's Hebbian emergence	Exact replication, both distance conventions	falsified (drive-covariance artifact)

My own Prop 4.2 protocol-dependence	Two independent protocols (stroboscopic, continuous-time)	residual corr varies (−0.07...0.60); obstruction invariant
ERM Wigner–Dyson claim	Both kernels; 120 clouds; bulk-trimmed spacings	$\langle r \rangle$ 0.51/0.50 — survives; note $N = 96$ is small
Spectral dimension ≈ 3	log-derivative of return probability, 30 time points	plateau visible but size-limited; downgrade to M

12.2 Honest limitations of this report

1. **Finite-size cage.** The strict identification is exact-grade at $N = 12$; the $N \rightarrow \infty$ asymptotic statement (circulant Yukawa limits) is standard but not proven here. Everything ERM-side is $N \leq 96$.
2. **Negative literature claims.** "Unstudied combination" (§9) is a search verdict, not a theorem; the self-referential bootstrap may live in a literature I did not reach (e.g. self-consistent reservoir adaptation).
3. **The bootstrap map's fixed-point classification** (§4.5) is empirical; a proof of "fixed points = projector mixtures" in full generality is missing (Prop 4.2 proves it only for the dephased-covariance limit).
4. **Convention sensitivity.** The repository mixes distance conventions (Table 1-1); I analyzed the gate-certified one. Under the absolute-distance convention, strict-kernel statements are numerically similar (Toeplitz vs circulant at $N = 12$ differs at the boundary) but not identical; I did not redo the full battery for the path graph.
5. **Confidence calibration.** **H** labels attach to exact or nine-digit-reproducible facts; probability assignments in §10–11 are structured judgment and should be read as such.

12.3 Final answer to the task's framing question

Does the mathematical core of FIN have a more natural interpretation as an adaptive information-propagation operator than as a physical field theory? **Yes — with a correction of terms.** The natural interpretation of the strict kernel is not yet "adaptive": it is a **propagation medium** (a normal-ordered discrete Green function), and "adaptive" is a well-posed upgrade with exactly one obstruction (Prop 4.2) and one trilemma (§7.2). The legacy kernel is not a propagation operator at all but a shell selector. The field-theoretic reading adds nothing to either object — every physical claim in the repository reduces to properties these operators already have, or fail to have, as graph-theoretic objects. The information-dynamics reading is therefore not merely more charitable; it is the only one that makes the object's actual mathematical content visible. Whether that content is worth developing is answered in §10: one honest paper, modest venue, no paradigm. [**H**]

Appendix A — Reproducibility of all numerical claims

Kernels. Eq. (1-1): $\alpha = 4 \ln 2$, $\omega = \pi/4$, $\phi = \pi/6$, $\beta = 0.01$; Eq. (1-2): $\omega = 0.18575$, $\phi = 0.1625$, $\beta = 1$, $\eta = 1.8$. Domain $\mathbb{Z}_{12}$, cyclic min-distance, zero diagonal (artifact `report_qw2118...json` flags: `matrix_symmetric=True`, eigenvalues reproduced to 9 digits).

Symbol formula. Eq. (2-1). Fits: least squares over $m = 0..6$ (curve_fit, maxfev 5×10^4). nRMSE as defined in §2.2. Thm 2.2 residual: $\|R\|_F / \|K\|_F = 0.0286$; $AG_{00} = 1.08836$ vs $|C| = 1.08960$.

Continuum transforms. 3D Fourier–Hankel by adaptive quadrature to $r = 2000$ (legacy: shell-peaked FT, sign changes; strict: Yukawa fit corr 0.070, power-law $|FT| \sim q^{-0.63}$ mid-range).

Learning experiments. Repo replication: Oja, lr=decay=0.01, 30k iters, drive $\cos(\pi i/4 + t) + \mathcal{N}(0, 0.5)$; results §4.4. Self-driven stroboscopic: lr=decay=0.02, 30k iters, flight time 0.7, random phase per step; corr $(K, C) = 1.0000$, corr $(K, K_{\text{strict}}) = -0.0697$; rank $8.21 \rightarrow 4.90$. Continuous-time: $e^{iK\Delta t}$, $\Delta t = 0.4$, lr 0.02, 4k iters; residual corr 0.60; rank $8.21 \rightarrow 3.57$. Pure Hebb: linear $\|K\|$ growth. BCM: lr 0.05, 8k iters; corr 0.156, rank 4.68. Bootstrap: $K \mapsto \Phi(C(K))$, 60 steps, occupation redrawn; rank oscillates 4.1–9.4.

ERM. Gaussian clouds $\mathcal{N}(0, 2^2 I_3)$; $N \in \{24, 48, 96\}$; 60–120 realizations. Spacings: unfolded per realization, 6 modes trimmed at each edge, pooled; $\langle r \rangle$ as in Table 6-1. Spectral dimension: $d_s(t) = -2 \partial_{\ln t} \ln P(t)$, $P(t) =$ mean return probability of $|K|$ -weighted combinatorial Laplacian heat kernel, 30 log-spaced $t \in [10^{-2}, 10^2]$.

Fisher. Eq. (7-1) by forward differences ($\varepsilon = 10^{-4}$, verified stable at 10^{-5}); null direction residual $|\lambda_{\min}| < 10^{-6}$. Alternate tuple residual 6.1×10^{-5} .

Filter gains. White-noise power gain $\sum \lambda_m^2/12 = 0.538$ (strict); Perron gain $1.6603^2 = 2.757$. Reservoir MC: tanh nonlinearity, input gain 0.1, 11 delays, 4000 steps: MC = 0.10.

All random seeds fixed (7, 11, 42, 1–5 depending on experiment); NumPy/SciPy implementations; no repository code was executed for any reported number.

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